

Mealvana Endurance - Dietitian Consultation Document

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Meeting Date: _____ Dietitian Name: _____ App Version: 1.0
(Pre-release)

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1. App Overview

What Mealvana Endurance Does

Mealvana Endurance is a personalized nutrition planning app for endurance athletes. It generates science-based nutrition recommendations for: - **Pre-workout** nutrition (what to eat before training/racing) - **During-workout** nutrition (fueling during activity) - **Post-workout** nutrition (recovery) - **Daily nutrition** (planned future feature)

Supported Sports

- Running (current)
- Cycling (in development)
- Swimming (in development)
- Triathlon (planned)

User Data We Collect

Data Point	Currently Used?	Notes
Age	Collected, not used	Passed to backend but no formula adjustment
Gender	Collected, not used	Male/Female/Other options
Weight	Yes	Core input for all calculations
Height	Collected, limited use	Body composition context
Gut Training Level	Yes	Low/Moderate/High - affects carb limits
Distance	Yes	Converted to km for calculations
Pace	Yes	Used to calculate MET/intensity
Duration	Yes	Calculated from distance/pace
Time Before Activity	Yes	Affects pre-workout recommendations
Temperature	Collected, not used	Passed but not in formulas
Humidity	Collected, not used	Passed but not in formulas
Sweat Rate Category	Collected, not used	Low/Medium/High options

Data Point	Currently Used?	Notes
Food Preferences	Yes	Liked/Disliked/Willing to try
Training History	Planned	Will have full training log

2. Current Algorithm Summary

2.1 Energy Expenditure Calculation

Current Formula (ACSM Running Equation):

Speed (mph) = 60 / pace (min/mile)

Speed (m/min) = speed (mph) × 26.8224

VO₂ = 0.2 × speed (m/min) + 3.5

MET = VO₂ / 3.5

Gross Calories = MET × weight (kg) × duration (hours)

Net Calories = ~1 kcal × weight (kg) × distance (km)

Example: 70kg athlete, 8:00/mile pace, 10 miles - Speed = 7.5 mph = 201.17 m/min - VO₂ = 43.73 ml/kg/min - MET = 12.5 - Duration = 80 min = 1.33 hours - Gross Calories = 12.5 × 70 × 1.33 = **1,166 kcal**

Dietitian Notes:

2.2 Pre-Workout Carbohydrates

Current Formula (Time-Based Scaling):

Time Before	Carbs (g/kg)	Example (70kg)
4+ hours	4.0	280g
3 hours	3.0	210g
2 hours	2.0	140g
1 hour	1.0	70g
15-60 min	0.5	35g
<15 min	0.25	17.5g

Pre-Workout Protein (Optional):

Protein (g) = $0.2 \times \text{weight (kg)}$
Example: 70kg → 14g protein

Pre-Workout Fat Cap:

If time > 2 hours: Fat cap = $0.2 \times \text{weight (kg)}$
If time ≤ 2 hours: Fat cap = $0.1 \times \text{weight (kg)}$

Dietitian Notes:

2.3 Pre-Workout Hydration

Current Formula:

Time Before	Fluid (ml/kg)	Example (70kg)
2+ hours	6.0	420ml
1-2 hours	5.0	350ml
<1 hour	3.0	210ml

Pre-Workout Sodium:

If time $\geq$ 2 hours: 400mg sodium
If time < 2 hours: 200mg sodium

Dietitian Notes:

2.4 During-Workout Carbohydrates

Current Formula:

Gut Training Multipliers: | Gut Training Level | Multiplier | Description | |—————|
 ————|—————| | Low | 0.7 | New to fueling during exercise | | Moderate | 0.8 | Some
 experience with during-workout nutrition | | High | 1.0 | Trained gut, experienced athlete |

Carb Rate Calculation:

```
Base Rate = Gut Multiplier × weight (kg)
Carb Rate (g/h) = CLAMP(Base Rate, 30, 60)
Total Carbs = Carb Rate × duration (hours)
```

Example: 70kg athlete, high gut training, 2-hour run - Base Rate = $1.0 \times 70 = 70$ g/h -
 Clamped Rate = 60 g/h (max cap) - Total = $60 \times 2 = \mathbf{120g \text{ carbs}}$

Dietitian Notes:

2.5 During-Workout Hydration

Current Formula (Intensity-Based):

Condition	Fluid Rate
Duration $\leq$ 1 hour	400 ml/h
MET $\geq$ 8 (high intensity)	750 ml/h
MET $\geq$ 6 (moderate)	600 ml/h
MET $<$ 6 (easy)	500 ml/h

During-Workout Sodium:

If duration $\leq$ 1 hour: 0 mg/h
If duration $>$ 1 hour: 250 mg/h

Dietitian Notes:

2.6 Post-Workout Recovery

Current Formulas:

Nutrient	Formula	Range	Example (70kg)
Carbs	$1.0 \times \text{weight (kg)}$	30-140g	70g
Protein	$0.3 \times \text{weight (kg)}$	18-45g	21g
Fluid	$1.5 \times \text{during-workout fluid}$	500-1200ml	Varies
Sodium	Fixed	500mg	500mg

Dietitian Notes:

3. Decision Points & Questions

3.1 Gender Adjustments

Research Finding: Women oxidize 10-56% more fat than men during endurance exercise, potentially requiring different carb recommendations. Women also have lower absolute sweat rates but similar relative dehydration risk.

Current Approach: No gender adjustment

Options for Discussion:

Option	Description	Multiplier
A	No adjustment (current)	1.0 for all
B	Slight reduction for women	0.9× carbs for women
C	Moderate reduction for women	0.85× carbs for women
D	Other	_____

Recommendation: _____

Should we adjust hydration/sodium for gender? - ☐ No adjustment needed - ☐

Women need less absolute fluid but same relative - ☐ Other:

Notes:

3.2 Age Adjustments

Research Finding: Masters athletes (35+) may need higher protein (1.8-2.2 g/kg/day vs 1.4-1.6 g/kg/day). Metabolic rate may decrease with age.

Current Approach: No age adjustment

Options for Discussion:

Protein Adjustments by Age: | Age Group | Protein Multiplier | Notes | |———|

—————|———| | <35 | 1.0× (baseline) | | 35-50 | **×** | | **50-65** | **×** | | 65+ | ____× | |

Carb/Hydration Adjustments by Age: - [] No adjustment needed - [] Adjustment needed: _____

Notes:

3.3 Weight-Based vs Fixed Sodium

Current Approach: Fixed sodium values (200-400mg pre, 250mg/h during, 500mg post)

Research Finding: Sodium loss varies 10-fold between individuals (500-2500+ mg/hour). Average is ~1,000-1,500 mg/hour during exercise.

Options for Discussion:

Option	Pre-Run	During (per hour)	Post-Run
A (Current)	200-400mg fixed	250mg/h fixed	500mg fixed
B	5mg/kg	10mg/kg/h	7mg/kg
C	400mg fixed	500mg/h fixed	750mg fixed
D	Based on sweat rate category	Based on sweat rate	Based on sweat rate
E	Other	Other	Other

If using sweat rate categories, what multipliers?

Sweat Rate	Multiplier
Low	___x
Medium	___x
High	___x

Recommendation: _____

Notes:

3.4 Environmental Adjustments (Temperature & Humidity)

Current Approach: Temperature and humidity are collected but NOT used in calculations

Research Finding: Hot/humid conditions can increase sweat rate 30-50%+. Heat acclimatization takes ~2 weeks.

Options for Discussion:

Temperature Adjustments: | Temperature | Fluid Multiplier | Sodium Multiplier | |—————|
 —————|—————| | <60°F / <15°C (Cool) | **x** / x | | 60-75°F / 15-24°C (Moderate) |
 1.0x (baseline) | 1.0x (baseline) | | 75-85°F / 24-29°C (Warm) | **x** / x | | >85°F / >29°C (Hot)
 | **x** / x |

Humidity Adjustments (additional multiplier): | Humidity | Additional Multiplier | |—————|
 —————| | <50% (Low) | **x** / | **50-70% (Moderate)** | **1.0x** / | **>70% (High)** | x |

Should temperature/humidity affect CARB recommendations? - [] No - [] Yes:

Notes:

3.5 Elevation/Altitude Adjustments

Current Approach: Not considered

Research Finding: Altitude increases energy expenditure and may affect hydration needs. ACSM formula assumes flat terrain.

Options for Discussion:

Should we adjust for elevation gain? | Elevation Gain | Energy Multiplier | |———|———|
|———|———| | Flat (<500ft / 150m) | 1.0× | | Moderate (500-2000ft / 150-600m) | **x** / /
Hilly (2000-4000ft / 600-1200m) | **x** | | Mountainous (>4000ft / >1200m) | ____× |

Should we adjust for training at altitude? | Altitude | Adjustment | |———|———|———|
<5000ft / <1500m | None | | 5000-8000ft / 1500-2400m | **x** / / **>8000ft / >2400m** | **x** |

Notes:

3.6 Intensity Zones (Beyond MET)

Current Approach: Use MET calculated from pace for hydration adjustments only

Research Finding: FATmax (maximum fat oxidation) occurs at 50-80% max HR individually. Below FATmax, less carbs needed. Above, more carbs critical.

Options for Discussion:

Should we use heart rate zones instead of/in addition to MET? | Zone | % Max HR | Carb Adjustment | Fluid Adjustment | |——|———|———|———|———| | Zone 1 (Recovery) | <60% | ✖ / ✖ | | Zone 2 (Aerobic) | 60-70% | ✖ / ✖ | | Zone 3 (Tempo) | 70-80% | 1.0x | 1.0x | | Zone 4 (Threshold) | 80-90% | ✖ / ✖ | | Zone 5 (VO2max) | >90% | ✖ / ✖ |

For Zone 2 / base training, should carb recommendations be reduced? - ☐ No, same recommendations - ☐ Yes, reduce by ____% - ☐ Yes, make optional / "train low" option

Notes:

3.7 Gut Training Progression

Current Approach: Three fixed levels (Low: 30g/h, Moderate: 45g/h, High: 60g/h)

Research Finding: Gut can be trained to absorb 90-120g/h with multiple transportable carbohydrates (glucose + fructose) over 2-6 weeks.

Options for Discussion:

Should we expand gut training levels? | Level | Max Carb Rate | Description | |———|
 ————|———| | Untrained | g/h | **Never fuels during exercise** | | **Low** | g/h |
 Occasional fueling | | Moderate | g/h | Regular fueling practice | | High | g/h |
 Experienced, single-source carbs | | Elite | g/h | Trained with glucose+fructose mix |

Should we provide gut training guidance/progression plans? - [] No, just recommendations - [] Yes, basic guidance - [] Yes, detailed week-by-week protocol

Notes:

3.8 Duration Thresholds

Current Approach: - <1 hour: 400ml/h fluid, 0mg sodium during - >1 hour: Full recommendations

Question: At what duration should athletes START fueling with carbs during exercise?

Option	Threshold	Notes
A	>45 minutes	
B	>60 minutes	
C	>75 minutes	
D	>90 minutes	
E	Intensity-dependent	

Recommendation: _____

Should short workouts (<60-90 min) be "fasted" or low-carb by default? - ☐ Yes, recommend minimal/no carbs - ☐ No, still provide recommendations but note optional - ☐ Depends on intensity: _____

Notes:

3.9 Training Period / Periodization

Current Approach: Not considered

Research Finding: Nutrition needs differ by training phase: - **Base phase:** Lower carb, fat adaptation focus - **Build phase:** Higher carb for intensity support - **Taper:** Glycogen loading - **Race day:** Maximum fueling

Options for Discussion:

Should we adjust recommendations by training phase?

Phase	Daily Carb (g/kg)	During-Workout Carb	Notes
Base / Off-season	___g/kg	___% of normal	
Build	___g/kg	___% of normal	
Peak / Race Week	___g/kg	___% of normal	
Taper	___g/kg	___% of normal	
Race Day	___g/kg	___% of normal	

Notes:

3.10 Training History / Cumulative Load

Current Approach: Not considered (each workout treated independently)

Research Finding: High training load from previous days increases glycogen depletion and recovery needs.

We will have access to full training logs. How should we adjust?

After high-volume day (e.g., long run yesterday): | Adjustment | Multiplier | |———|
 ———| | Pre-workout carbs | $\times$ | | ***During-workout carbs*** | $\times$ | | Post-workout carbs | $\times$ | |
Post-workout protein | $\times$ |

During high-volume training block: - Daily carb recommendation: g/kg/day - Daily protein recommendation: g/kg/day

Notes:

3.11 Sport-Specific Adjustments

Current Approach: Running only, no sport-specific formulas

Question: How should we adjust for different sports?

Cycling vs Running (same duration/intensity): | Factor | Running | Cycling | Swimming | |
——|——|——|——-| | Carb needs | 1.0x | $\times / \times$ | | Fluid needs | 1.0x | $\times / \times$ | | Sodium
needs | 1.0x | $\times / \times$ | | GI tolerance | Baseline | $\times / \times$ |

Sport-specific considerations: - Cycling: Can carry more, easier to eat while moving -
Swimming: Pool vs open water, harder to fuel during - Triathlon: Transition nutrition?

Notes:

3.12 Menstrual Cycle Considerations

Current Approach: Not tracked or considered

Research Finding: - Luteal phase: Higher protein needs (1.8-2.2 g/kg), tighter recovery window (30-60 min), possible carb increase for high-intensity - Progesterone affects sodium retention - Hyponatremia risk may increase in luteal phase

Should we track menstrual cycle phase? - ☐ No, too complex - ☐ Yes, optional feature - ☐ Yes, recommended feature

If yes, what adjustments?

Phase	Carb Adjustment	Protein Adjustment	Sodium Adjustment
Follicular (Day 1-14)	____x	____x	____x
Luteal (Day 15-28)	____x	____x	____x
Menstruation	____x	____x	____x

Notes:

4. Concrete Scenarios for Review

Please review each scenario and note whether our current recommendation is appropriate, too high, too low, or needs adjustment.

Scenario 1: Short Easy Run - Should They Eat?

Athlete: 35-year-old male, 70kg (154 lbs) **Activity:** 5-mile easy run, 9:00/mile pace (45 minutes) **Conditions:** 65°F, moderate humidity **Time before run:** 2 hours **Gut training:** Moderate

Our Current Recommendation: - Pre-run: 140g carbs, 350ml fluid, 400mg sodium - During: 0g carbs (under 1 hour), 300ml fluid, 0mg sodium - Post-run: 70g carbs, 21g protein, 450ml fluid, 500mg sodium

Assessment: - ☐ Appropriate - ☐ Pre-run too high - ☐ Pre-run too low - ☐ During recommendations need adjustment - ☐ Post-run too high - ☐ Post-run too low

What would you recommend?

Pre-run: _____

During: _____

Post-run: _____

Scenario 2: Long Run - Hot Weather

Athlete: 28-year-old female, 58kg (128 lbs) **Activity:** 18-mile long run, 8:30/mile pace (2.5 hours) **Conditions:** 85°F, 80% humidity **Time before run:** 3 hours **Gut training:** High

Our Current Recommendation: - Pre-run: 174g carbs, 348ml fluid, 400mg sodium - During: 150g carbs (60g/h × 2.5h), 1875ml fluid (750ml/h), 375mg sodium - Post-run: 58g carbs, 17g protein, 1125ml fluid, 500mg sodium

Assessment: - ☐ Appropriate - ☐ Pre-run needs adjustment: _____
 - ☐ During carbs too high/low: _____ - ☐ During fluids too high/low: _____
 - ☐ During sodium too high/low: _____ - ☐ Post-run needs adjustment: _____

What would you recommend given the heat/humidity?

Scenario 3: Masters Athlete - Marathon Training

Athlete: 55-year-old male, 82kg (180 lbs) **Activity:** 20-mile marathon pace run, 8:00/mile pace (2h 40min) **Conditions:** 70°F, low humidity **Time before run:** 4 hours **Gut training:** Moderate

Our Current Recommendation: - Pre-run: 328g carbs, 492ml fluid, 400mg sodium - During: 128g carbs (48g/h capped from 65.6), 1600ml fluid, 400mg sodium - Post-run: 82g carbs, 25g protein, 1200ml fluid, 500mg sodium

Assessment: - ☐ Appropriate for a 55-year-old - ☐ Needs age-specific adjustments: _____ - ☐ Protein recommendation should be higher - ☐ Other: _____

Age-specific recommendations?

Scenario 4: Ultra-Distance - Hyponatremia Risk

Athlete: 42-year-old female, 63kg (139 lbs) **Activity:** 50-mile ultra, 12:00/mile pace (10 hours) **Conditions:** 75°F, moderate humidity **Time before run:** 4 hours **Gut training:** High

Our Current Recommendation: - Pre-run: 252g carbs, 378ml fluid, 400mg sodium - During: 600g carbs (60g/h × 10h), 6000ml fluid (600ml/h), 2250mg sodium - Post-run: 63g carbs, 19g protein, 1200ml fluid, 500mg sodium

Key Concerns: 1. Is 600g carbs during realistic/safe? 2. Is 6 liters of fluid appropriate or hyponatremia risk? 3. Is 2250mg sodium sufficient for 10 hours? 4. Female athlete - any specific concerns?

Assessment:

Carbs: _____
Fluids: _____
Sodium: _____
Other: _____

Scenario 5: Zone 2 Base Training

Athlete: 30-year-old male, 75kg (165 lbs) **Activity:** 90-minute easy run, 10:00/mile pace, Zone 2 heart rate **Conditions:** 60°F, low humidity **Time before run:** 1 hour **Gut training:** Moderate **Goal:** Fat adaptation / base building phase

Our Current Recommendation: - Pre-run: 75g carbs, 375ml fluid, 200mg sodium - During: 60g carbs (40g/h × 1.5h), 750ml fluid, 125mg sodium - Post-run: 75g carbs, 22g protein, 750ml fluid, 500mg sodium

Question: For base training / fat adaptation goals, should carbs be reduced?

Assessment: - ☐ Keep current recommendations - ☐ Reduce pre-run carbs to: _____
- ☐ Reduce during carbs to: _____ - ☐ Make fueling optional for Zone 2 - ☐ Other:

Notes:

Scenario 6: Fasted Morning Run

Athlete: 38-year-old female, 60kg (132 lbs) **Activity:** 6-mile tempo run, 7:30/mile pace (45 minutes), Zone 4 **Conditions:** 55°F, low humidity **Time before run:** 15 minutes (just woke up) **Gut training:** High

Our Current Recommendation: - Pre-run: 15g carbs, 180ml fluid, 200mg sodium - During: 0g carbs, 300ml fluid, 0mg sodium - Post-run: 60g carbs, 18g protein, 450ml fluid, 500mg sodium

Question: Is fasted high-intensity running advisable? What minimum should we recommend?

Assessment: - ☐ Current recommendations OK - ☐ Should NOT recommend fasted for tempo/high intensity - ☐ Minimum pre-run: _____ g carbs - ☐ Other:

Notes:

Scenario 7: Back-to-Back Training Days

Athlete: 26-year-old male, 68kg (150 lbs) **Yesterday:** 22-mile long run **Today's Activity:** 8-mile recovery run, 9:30/mile pace (1h 16min) **Conditions:** 70°F, moderate humidity
Time before run: 2 hours **Gut training:** Moderate

Our Current Recommendation (same as if no previous long run): - Pre-run: 136g carbs, 340ml fluid, 400mg sodium - During: 41g carbs, 608ml fluid, 125mg sodium - Post-run: 68g carbs, 20g protein, 912ml fluid, 500mg sodium

Question: After a 22-mile run yesterday, should we increase recommendations?

Adjustment recommendations:

Pre-run carbs: _____

Other adjustments: _____

Scenario 8: Heavy Sweater in Heat

Athlete: 45-year-old male, 90kg (198 lbs) **Activity:** Half marathon, 7:00/mile pace (1h 32min) **Conditions:** 88°F, 85% humidity **Sweat rate:** Self-reported "heavy sweater" **Time before run:** 3 hours **Gut training:** High

Our Current Recommendation: - Pre-run: 270g carbs, 540ml fluid, 400mg sodium - During: 92g carbs, 1150ml fluid (750ml/h), 250mg sodium - Post-run: 90g carbs, 27g protein, 1200ml fluid, 500mg sodium

Specific Concerns: 1. Heavy sweater + extreme heat - is sodium sufficient? 2. Should we recommend sodium loading before? 3. What fluid rate for someone who sweats heavily?

Assessment:

Sodium pre: _____
Sodium during: _____
Fluid rate: _____
Other: _____

Scenario 9: Type 2 Diabetic Athlete

Athlete: 52-year-old male, 85kg (187 lbs), Type 2 diabetic (managed with medication)

Activity: 10-mile run, 10:00/mile pace (1h 40min) **Conditions:** 72°F, moderate humidity

Time before run: 2 hours **Gut training:** Low

Our Current Recommendation (standard): - Pre-run: 170g carbs, 510ml fluid, 400mg sodium - During: 58g carbs (35g/h due to low gut training), 900ml fluid, 167mg sodium - Post-run: 85g carbs, 26g protein, 1200ml fluid, 500mg sodium

Questions: 1. Should we reduce carb recommendations for diabetic athletes? 2. Any specific carb types to recommend/avoid? 3. Any safety warnings we should display?

Assessment:

Scenario 10: Keto-Adapted Athlete

Athlete: 40-year-old female, 62kg (137 lbs), follows ketogenic diet **Activity:** 3-hour trail run, easy pace **Conditions:** 65°F, low humidity **Time before run:** 3 hours **Gut training:** N/A (doesn't train with carbs)

Our Current Recommendation (standard carb-based): - Pre-run: 186g carbs, 372ml fluid, 400mg sodium - During: 180g carbs, 1500ml fluid, 500mg sodium - Post-run: 62g carbs, 19g protein, 1200ml fluid, 500mg sodium

Questions: 1. Should we have a "keto mode" with different recommendations? 2. What should we recommend for keto athletes? 3. Any warnings about intensity limitations?

Assessment:

Scenario 11: Cyclist - Same Energy Expenditure

Athlete: 33-year-old male, 72kg (159 lbs) **Activity:** 3-hour bike ride, moderate intensity (similar calories to 2h run) **Conditions:** 75°F, moderate humidity **Time before ride:** 2 hours **Gut training:** High

Questions: 1. Same recommendations as equivalent run? 2. Higher carb tolerance on bike (less GI jostling)? 3. Different hydration needs?

Running equivalent recommendation: - During: 180g carbs, 1800ml fluid, 500mg sodium

What should cycling be?

During carbs: _____

During fluids: _____

During sodium: _____

Other differences: _____

Scenario 12: Sprint Triathlon

Athlete: 29-year-old female, 57kg (125 lbs) **Activity:** Sprint triathlon (750m swim, 20km bike, 5km run) - ~1.5 hours total **Conditions:** 78°F, moderate humidity **Time before race:** 2 hours **Gut training:** Moderate

Questions: 1. Should we break down by discipline or treat as single event? 2. Transition nutrition recommendations? 3. When to fuel during such a short multi-sport event?

Assessment:

Pre-race: _____

During swim: _____

T1 (swim to bike): _____

During bike: _____

T2 (bike to run): _____

During run: _____

Post-race: _____

Scenario 13: Young Athlete

Athlete: 17-year-old female, 52kg (115 lbs) **Activity:** 10K cross country race, 6:30/mile pace (40 minutes) **Conditions:** 55°F, low humidity **Time before race:** 3 hours **Gut training:** Low

Our Current Recommendation: - Pre-race: 156g carbs, 312ml fluid, 400mg sodium - During: 0g carbs (under 1 hour), 267ml fluid, 0mg sodium - Post-race: 52g carbs, 16g protein, 400ml fluid, 500mg sodium

Questions: 1. Any age-specific considerations for teen athletes? 2. Different minimum age requirements for certain recommendations? 3. RED-S concerns for young female athletes?

Assessment:

Scenario 14: Luteal Phase Female Athlete

Athlete: 31-year-old female, 60kg (132 lbs), currently in luteal phase (day 20 of cycle)

Activity: Track workout - 6 × 800m with recoveries (75 minutes total) **Conditions:** 70°F, moderate humidity **Time before workout:** 1.5 hours **Gut training:** High

Our Current Recommendation (no cycle adjustment): - Pre-workout: 90g carbs, 330ml fluid, 200mg sodium - During: 56g carbs, 750ml fluid, 125mg sodium - Post-workout: 60g carbs, 18g protein, 750ml fluid, 500mg sodium

Questions: 1. Should carbs/protein be adjusted in luteal phase? 2. Any sodium considerations (progesterone effects)? 3. Is recovery timing more critical in luteal phase?

Assessment:

Carb adjustment: _____

Protein adjustment: _____

Sodium adjustment: _____

Recovery window: _____

Scenario 15: Race Day vs Training Day

Athlete: 36-year-old male, 78kg (172 lbs) **Activity:** Marathon - 26.2 miles, 7:15/mile pace (3h 10min) **Conditions:** 58°F, low humidity (ideal) **Time before race:** 3.5 hours **Gut training:** High **Context:** This is race day, not a training run

Our Current Recommendation (same as training): - Pre-race: 273g carbs, 468ml fluid, 400mg sodium - During: 190g carbs (60g/h), 1900ml fluid, 525mg sodium - Post-race: 78g carbs, 23g protein, 1200ml fluid, 500mg sodium

Questions: 1. Should race day have different (higher?) recommendations than training? 2. Carb loading protocol for days before? 3. Any race-specific strategies?

Assessment:

Race day differences: _____

Pre-race loading (days before): _____

Other: _____

5. Safety Considerations

5.1 Hyponatremia Risk

Current Safeguards: We limit during-workout sodium to 250mg/h regardless of duration.

Questions: 1. At what duration does hyponatremia become a real concern? _____ 2. What sodium/fluid ratio prevents hyponatremia? _____ 3. Should we warn users not to drink beyond thirst for ultras? _____ 4. Maximum fluid intake warning (ml/hour)? _____

Warning thresholds to implement:

5.2 Hypoglycemia (Diabetic Athletes)

Current Safeguards: None specific

Questions: 1. What warnings should we show for diabetic athletes? 2. Should we recommend different timing for insulin-dependent athletes? 3. What minimum carb intake should we enforce?

Recommendations:

5.3 RED-S (Relative Energy Deficiency in Sport)

Current Safeguards: None

Questions: 1. Should we track energy availability? 2. What warning signs should trigger alerts? 3. Minimum calorie thresholds by activity?

Recommendations:

5.4 GI Distress Prevention

Current Safeguards: Gut training levels limit carb intake, phase-appropriate foods only

Questions: 1. What carb rate reliably causes GI issues? _____ g/h 2. Should we warn against high-fiber foods pre-run? At what timing? 3. Fat limits before exercise? _____g at what timing? 4. Protein limits before exercise? _____g at what timing?

Recommendations:

5.5 Under-Fueling Warnings

Questions: 1. At what duration/intensity should we warn if user skips fueling? 2. Minimum carb intake for races over 90 minutes? _____g/h 3. When is fasted training contraindicated?

Recommendations:

6. Controversial Topics

6.1 Fasted Training

Research shows: Mixed evidence. May improve fat oxidation but no studies show actual performance improvement. May compromise high-intensity training.

Our current position: We provide recommendations even for short pre-run windows but don't actively encourage or discourage fasting.

Questions: 1. When is fasted training appropriate (if ever)? 2. When should we actively discourage it? 3. Should we have a "fasted training mode" with different recommendations?

Your recommendation:

6.2 Ketogenic Diets for Endurance

Research shows: May work for ultra-endurance at low intensity but compromises performance above ~80% VO₂max. ISSN recommends against for most athletes.

Our current position: Standard carb-based recommendations only.

Questions: 1. Should we support keto athletes with alternative recommendations? 2. If yes, what would those look like? 3. What warnings should we provide?

Your recommendation:

6.3 "Train Low, Compete High"

Research shows: 73-78% of studies show metabolic adaptations but compromises high-intensity training. Some experts now call this "dated advice."

Our current position: Not implemented.

Questions: 1. Should we offer this as an option for advanced users? 2. What guardrails would you recommend? 3. Or should we avoid this entirely?

Your recommendation:

6.4 Carb Intake Above 60g/h

Research shows: With glucose+fructose (multiple transportable carbs), athletes can absorb 90-120g/h after gut training. But gut training is required.

Our current maximum: 60g/h (hard cap)

Questions: 1. Should we allow higher rates for trained athletes? 2. What criteria should unlock 90g/h? 120g/h? 3. What glucose:fructose ratio should we recommend?

Your recommendation:

6.5 Sodium Loading Before Events

Research shows: 1,500mg sodium in 32oz water before events may improve performance in heat.

Our current position: Fixed 400mg pre-workout sodium.

Questions: 1. Should we recommend sodium loading for certain conditions? 2. What criteria? (Duration, temperature, sweat rate?) 3. What amount/timing?

Your recommendation:

7. Additional Notes

7.1 Daily Nutrition Guidance (Future Feature)

We plan to add daily nutrition guidance. Initial questions:

1. Daily carb recommendations by training load?

- Light training day: _____g/kg
- Moderate training day: _____g/kg
- Heavy training day: _____g/kg
- Rest day: _____g/kg

2. Daily protein recommendations?

- Standard athlete: _____g/kg
- Masters athlete (35+): _____g/kg
- High volume training: _____g/kg

3. Should daily recommendations account for workout nutrition?

- ☐ Yes, subtract workout intake from daily total
- ☐ No, treat separately
- ☐ Other: _____

Notes:

7.2 Any Other Factors We Should Consider?

7.3 Resources / References Recommended

7.4 Follow-Up Questions for Future Meetings

Meeting Summary

Key Decisions Made:

1.

2.

3.

4.

5.

Action Items:

1.

2.

3.

4.

5.

Follow-Up Meeting Needed? ☐ Yes ☐ No

Next Meeting Date: _____

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